Independent audit of the FINAL_V1 experiment

Audit by Opus 5.5 (max effort): bugs, glitches, fixes and verification evidence

Protocol FINAL_V1 Status: **COMPLETED** Generated: 2026-10-20T13:09:03Z

Verdict

- No critical bug; no reported prediction or metric changed.
- 2 major documentation errors and 8 minor glitches found and fixed; 2 informational notes. Frozen protocol and modelling code left byte-identical; corrections recorded in PROTOCOL_ERRATA.md and DECISION_LOG.md (D-009).
- All 20 OOF-audit checks and all 25 pipeline tests pass after the fixes.

Verified without findings

- Data identity and labels: Independent re-parse of the raw tree: 785/785 files matched, 0 label mismatches, 47 speakers, identical partition: True.
- Duplicates: No cross-speaker recording pair with WavLM cosine ¿= 0.999 (max 0.9948; within-speaker max 0.9976). One byte-identical duplicate already excluded.
- Encoders: Load reports list only unused pre-training heads (no missing weights). Fresh-process re-extraction of 5 recordings per encoder: max —difference— 3.1e-06, cosine 1.0.
- Acoustic features: Independent implementation (25 recordings x 40 features): max relative difference 9.6e-06.
- Nested CV: Independent re-implementation for WavLM-LR: max —dp— 1.3e-14, same configuration in 5/5 folds; XLS-R-LR: 3.9e-04 (BLAS threading), 5/5 folds.
- Metrics: All 56 conditions x 6 metrics recomputed from prediction files and compared with condition_metrics.csv and the registry: 0 disagreements.
- Bootstrap: Percentile intervals and paired differences recomputed with independent code for two comparisons: identical to 1e-12.
- Thread sensitivity: 70 LR runs re-executed with single-threaded BLAS: max —dp— 4.4e-16; same configurations in all folds; identical pooled metrics.
- OOF audit: 20 checks, all passed after the fixes (280 runs, 0 failed).
- Isolation: Original project unchanged since the start snapshot; 831 original audio files re-hashed, 0 changed.

Table 1: Findings and fixes

ID	Severity	Finding	Fix	Result changed?
AUD-01	MAJOR (documentation)	The two-digit number in the file names was described as age (protocol D-003 text, DATA_AUDIT, DECISION_LOG, stage_a_build docstring, error analysis of the guideline report).	Errata E-1 (PROTOCOL_ERRATA.md); documents and reports corrected; sheet demographics added.	No
AUD-02	MAJOR (data documentation)	Spreadsheet notes on four starred controls (poor-quality recordings; no passage/no phrases; reading difficulty) were not reported.	Errata E-2; DATA_AUDIT, FINAL_REPORT and guideline report updated.	No
AUD-03	MINOR (reporting)	Embedding extraction stated as 'roughly 1.5 hours'; summed run times called 'core-hours'.	FINAL_REPORT.md/.pdf and the Stage D PDF corrected.	No
AUD-04	MINOR (reporting)	Guideline report wording: 'different equipment' (not documented), 'which is why extensions cannot improve' (causal), encoders '315-317M' parameters, 'about 630' training recordings per fold, '41 to 3,113 parameters' including XGBoost.	Wording corrected; reports regenerated.	No
AUD-05	MINOR (execution harness)	Runner gate 'protocol_and_split_hashes_match_freeze' compared the freeze record with itself (always true).	Gate now hashes the on-disk files; the OOF audit had verified the on-disk hashes independently (check 7). Stage C tests re-run: 25/25 pass.	No
AUD-06	MINOR (reproducibility scope)	'Bit-identical replay' was shown for 15 runs only.	Documented in FINAL_REPORT (14b), ENVIRONMENT.md, guideline report.	No
AUD-07	MINOR (code hygiene)	pipeline.py silences all warnings.	Not changed (frozen modelling code); documented.	No
AUD-08	MINOR (statistics presentation)	Percentile bootstrap intervals collapse to [1.000, 1.000] when the pooled AUROC is 1.	Noted in FINAL_REPORT; exact Clopper-Pearson intervals for sensitivity/specificity added in the whitepaper.	No
AUD-09	MINOR (compliance)	FAILURE_LOG.csv was empty although development failures and the FWL block occurred.	7 rows added (all RESOLVED except FWL BLOCKED).	No
AUD-10	MINOR (presentation)	Feature-selection stability table repeats identical rows for the four models.	Explanatory note added to the PDF.	No
AUD-11	INFO	Two runner gates (labels, folds) are consistency checks by construction.	None needed.	No
AUD-12	INFO	ENVIRONMENT.md / REPRODUCIBILITY.md incomplete (torch build, packages added later, full command list).	Both updated.	No